Chapter 3
SPNU514C–March 2018

# Power Management Module (PMM)

This chapter describes the power management module (PMM).

**Topic** **Page**

## 3.1 Overview

This microcontroller is part of the Hercules family of microcontrollers from Texas Instruments for safety-critical applications. Several functions are implemented on this microcontroller targeted towards varied applications. The core logic is divided into several domains that can be independently turned on or off based on the application's requirements. This allows an application to reduce the leakage current for a core domain that has modules that are not being used by the application.

This chapter describes the Power Management Module (PMM). The PMM provides memory-mapped registers that control the states of the supported power domains. The PMM includes interfaces to the Power Mode Controller (PMC) and the Power State Controller (PSCON). The PMC and PSCON control the power up/down sequence of each power domain.

### *3.1.1 Main Features of the Power Management Module (PMM)*

The main features of the PMM implemented on the microcontroller are:

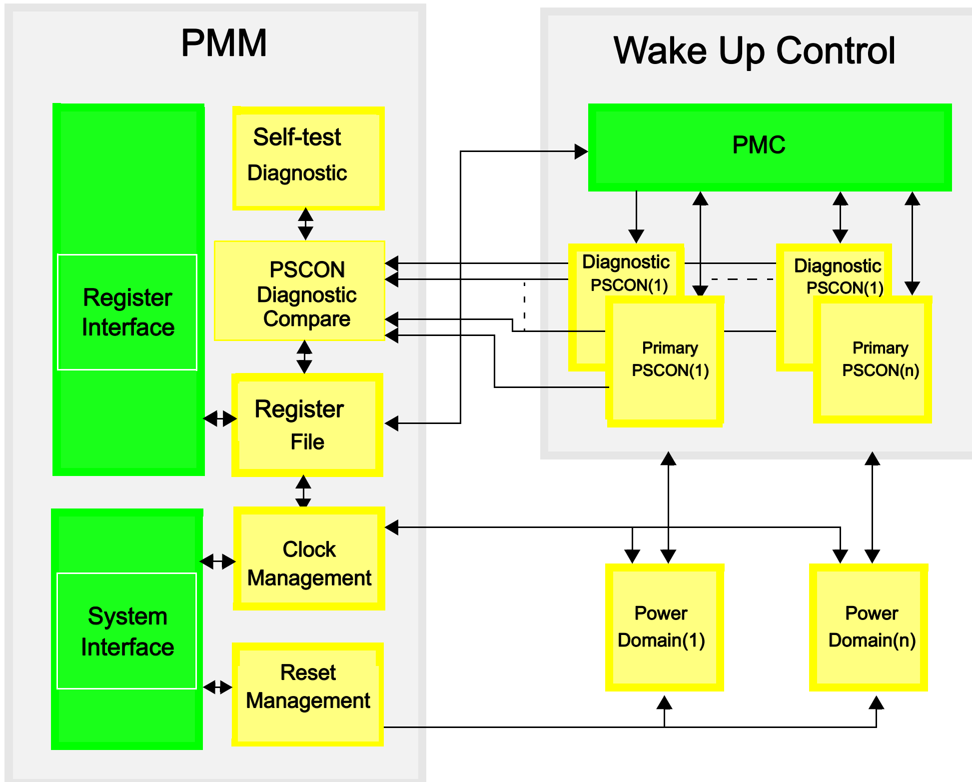
- Supports 5 logic power domains: PD1, PD2, PD3, PD4, and PD5
- Supports 2 memory-only power domains: RAM_PD1, RAM_PD2
- Manages the clocks for each power domain
- Manages the resets to each power domain
- Includes failsafe compare logic to continuously monitor the states of each power domain
- Supports diagnostic and self-test logic to validate failsafe compare logic

### *3.1.2 Block Diagram*

Figure 3-1 is a block diagram of the PMM. The PMM consists of several key components:

- Register interface – the PMM control registers are mapped to the device memory space and start at address 0xFFFF0000.
- System Interface – the PMM receives the clocks, resets, errors and all other control signals through this interface.
- PSCON Diagnostic Compare – this block compares the outputs of each primary PSCON and the respective diagnostic PSCON implemented for failsafe safety.
- Self-Test Diagnostic – this block contains the logic to place the PSCON diagnostic compare block in a self-test mode in order to test the failsafe feature.
- Clock management – the PMM provides independent clock gating and handshaking controls for each power domain and also generates the clock domains for each power domain.
- Reset Management – the PMM provides independent reset signals for each power domain.
- Power State Controller (PSCON) – The PSCON is a finite state machine that controls the power sequence of a power domain from one state to another. Each power domain is controlled by one dedicated PSCON.
- Power Domain – A power domain is a group of logic and/or memories which is separated from the global power supply via power switches. These power switches are controlled by the PSCON and can be turned on or off.

**Figure 3-1. PMM Block Diagram**



## 3.2 Power Domains

Figure 3-2 shows the core and memory power domains implemented on the microcontroller.

This device has 7 separate core power domains:

- PD1 is an always-ON domain and is not controlled by PMM. It contains the CPU as well as other principal modules and the interconnect required for operation of the microcontroller. This domain also includes 64KB of the tightly-coupled RAM. The PD1 can operate on its own even when all the other core power domains are turned off by the PMM. Note that all I/Os are in this always-ON domain as well.

Core power domains PD2 through PD5 and RAM_PD1 through RAM_PD2 are controlled by the PMM.

- PD2 contains the Parameter Overlay Module (POM) and test logic.
- PD3 contains some additional peripheral modules as an enhanced configuration over and above the peripheral set available in PD1. These include a second High-End Timer (N2HET2) with its dedicated transfer unit (HTU2), a second Analog-to-Digital Converter (ADC2), a Serial Communication Interface (SCI), an Inter-Integrated Circuit controller (I2C), a third Controller Area Network controller (DCAN3), and a fourth Serial Peripheral Interface module (SPI4).
- PD4 is unavailable in this microcontroller.
- PD5 contains the Ethernet Controller (EMAC), External Memory Interface (EMIF), Enhanced Pulse Width Modulator (ePWMx), Enhanced Capture (eCAPx), and Enhanced Quadrature Encoder Pulse (eQEPx), as well as some components of the interconnect fabric required by these modules.
- RAM_PD1 and RAM_PD2 each contain 64KB of tightly-coupled RAM.

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**NOTE: Switching of Power Domains**

The microcontrollers only support static switching of the power domains. That is, the power domains can be turned ON or OFF one time during device initialization. Once configured, it is not allowed to change the state of a power domain without first asserting a system reset.

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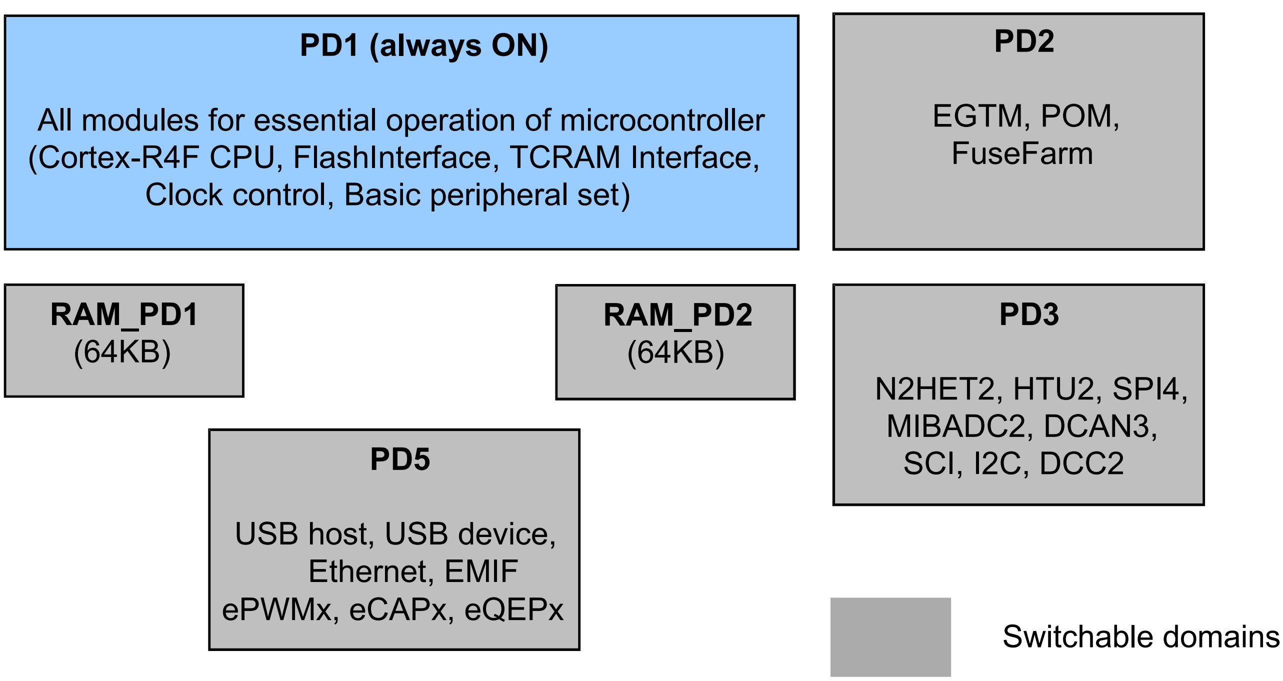
**Figure 3-2. Core Power Domains**



## 3.3 PMM Operation

It is important to understand some fundamental concepts beforehand.

### *3.3.1 Power Switch*

A power domain gets its power supply via a power switch. The power switch creates a link between the global core supply plane and the local switchable power domain supply. Each power domain uses multiple power switches, which are daisy-chained together.

### *3.3.2 Power Domain State*

Each core power domain can be in one of three states: Active, Idle, or Off.

In the ***active*** state, a power domain is fully powered with normal supply voltage.

In the ***idle*** state, all clocks to a power domain are turned off (driven low). The supply voltage is still maintained at the normal level.

In the ***off*** state, a power domain is completely cut off from the power supply.

### *3.3.3 Default Power Domain State*

The default state of each power domain, except for PD1, is controlled by TI during production testing via programmation of individual bits within the reset configuration word in the TI-OTP sector of flash bank 0. This allows each power domain to default to either the active state or the off state.

### *3.3.4 Disabling a Power Domain Permanently*

TI can also permanently disable any power domain, except for PD1. This is also controlled by programmation of individual bits within the reset configuration word in the TI-OTP sector of flash bank 0.

### *3.3.5 Changing Power Domain State*

A domain can only change state when commanded by the application. Each domain has an associated 4-bit key to define the intended power state. When the correct key is programmed, the PMM initiates the sequence to transition that domain to the commanded state.

A power state transition is considered complete only when every single power switch for that domain has switched over to the commanded state.

#### 3.3.5.1 Turning a Power Domain Off

It is necessary to turn off all clocks going to a power domain before that domain can be powered down. PMM contains the hardware interlocks to handle this. Each power domain has an associated memory-mapped register which allows the application to turn off clocks to that power domain.

Steps to power down a domain with logic – PD2, PD3, PD4, PD5:

1. Write to the PDCLK_DISx register to disable all clocks to the power domain.
2. Write 0xA to the LOGICPDPWRCTRL0 register to power down each domain.
3. Poll for LOGICPDPWRSTATx.LOGICPDPWR_STATx.to become “00”. The power domain is now powered down.

A power domain with only SRAM macros does not have a clock input, so the sequence is shorter. This applies to RAM_PD1 and RAM_PD2 power domains shown in Figure 3-2:

1. Write the correct key to the MEMPDONx register to power down the domain.
2. Poll for MEMPDPWRSTATx.MEMPDPWR_STATx to become “00”. The power domain is now powered down.

#### 3.3.5.2 Turning a Power Domain On

A power domain can be turned on by writing the correct key to the LOGICPDPWRCTRL0.LOGICPDONx or MEMPDPWRCTRL0.MEMPDONx. PMM will automatically restart the clocks to the power domain once the power is restored if the "automatic clock enable upon wake up" option is selected. If this option is not selected, the application can turn on clocks to the power domain by clearing the PDCLK_DIS register. The application must poll the LOGICPDPWRSTATx.DOMAIN_ONx to ensure that the power has been fully restored before enabling the clocks.

**NOTE:** If a power domain has been permanently disabled by TI, the application cannot turn this power domain ON. No error is generated if the application attempts to do so.

### *3.3.6 Reset Management*

PMM handles the reset sequence for each power domain. When a power domain is turned on from an off state, the PMM will reset the power domain to ensure that all logic begins in its default reset state.

PMM generates nPORRST (power-on reset), nRST (system reset), nPRST (peripheral reset), and nTRST (test / debug logic reset) for each domain.

### *3.3.7 Diagnostic Power State Controller (PSCON)*

Each power domain state is controlled by a primary PSCON. There is a second PSCON as well for each power domain. This is the diagnostic PSCON. All power management inputs to a power domain are controlled only by the primary PSCON. All power management outputs from the power domain are fed back to both the primary and the diagnostic PSCON.

The PMM commands both the PSCON identically so that they are always in a lock-step operating mode. A dedicated compare unit checks the outputs of the two PSCON modules on every cycle.

### *3.3.8 PSCON Compare Block*

The diagnostic compare block can operate in one of four modes:

#### 3.3.8.1 Lock-Step Mode

This is the default mode of operation of the PSCON compare block. The PSCON diagnostic compare block compares the outputs from the two PSCONs on every cycle. Any mismatch in the PSCON outputs is indicated as a PSCON compare error. This error signal is mapped to the Error Signaling Module's (ESM) Group1 channel 38. The application can define the response to this error.

#### 3.3.8.2 Self-Test Mode

A self-test mechanism is provided to check the PSCON compare logic for faults. The compare error signal output is disabled in self-test mode. The PSCON diagnostic compare block generates two types of patterns during self-test mode: compare match test followed by compare mismatch test. During the self-test, each test pattern is applied on both PSCON signal ports of the PSCON diagnostic compare block and then is clocked for one cycle. The duration of the self-test is 24 cycles. Any detected fault is indicated as a self-test error, mapped to ESM group1 channel 39. If no fault is detected, the self-test complete flag is set.

The application can poll for this flag to be set and then switch the mode of the PSCON compare block back to lock-step mode by writing to the mode key register.

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**NOTE: PSCON operation when compare block is in self-test mode**

When the PSCON compare block is in its self-test mode, both PSCONs continue to function normally. However, there is no comparison done on the PSCON outputs.

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**Compare match test:**

An identical vector is applied to both input ports at the same time, thereby expecting a compare match. If the compare unit produces a mismatch then the self-test error flag is set and the self-test error signal is generated. The compare match test is terminated if a compare mismatch is detected. The compare match test takes 4 cycles to complete when the test passes.

**Compare mismatch test:**

A vector with all 1's is applied to the PSCON diagnostic compare block's primary input port and the same input is also applied to the secondary input port but with one bit flipped starting from bit position 0. The unequal vectors should cause the PSCON diagnostic compare block to generate a compare mismatch at bit position 0. In case a mismatch is not detected, a self-test error is indicated. This compare mismatch test algorithm is repeated until every single bit position is verified on both PSCON signal ports.

#### 3.3.8.3 Error-Forcing Mode

This mode is designed specifically to ensure that the error signal output from the PSCON compare block is not stuck inactive. In this mode, a test pattern is applied to the PSCON related inputs of the compare logic to force an error. During error forcing mode, both the error signal and the self-test error signal will be asserted to the ESM. The application can clear flags for ESM group1 channel 38 and ESM group1 channel 39 once the error is flagged. If the two ESM flags do not get set, this indicates that the PSCON compare error signal is stuck inactive and cannot be relied upon to detect a PSCON mismatch.

#### 3.3.8.4 Self-Test Error-Forcing Mode

In this mode, an error is forced so that the self-test error output from the PSCON compare block is activated. The application can clear the flag for ESM group1 channel 39 once the error is flagged. If the ESM group1 channel 39 flag does not get set, this indicates that the PSCON compare block self-test error signal is stuck inactive and there is no self-test mechanism available for the PSCON compare block.

#### 3.3.8.5 PMM Operation During CPU Halt Debug Mode

No compare errors are generated when the CPU is halted in debug mode, regardless of the mode of the diagnostic compare block. No status flags are updated in this mode. Normal operation of the compare block is resumed once the CPU exits the debug mode.

## 3.4 PMM Registers

Table 3-1 lists the memory-mapped registers for the PMM. All register offset addresses not listed in Table 3-1 should be considered as reserved locations and the register contents should not be modified. The address offset is specified from the base address of FFFF 0000h. Any access to an unimplemented location within the PMM register frame will generate a bus error that results in an Abort exception.

**Table 3-1. PMM Registers**

| Offset | Acronym | Register Name | Section |
|---|---|---|---|
| 0h | LOGICPDPWRCTRL0 | Logic Power Domain Control Register 0 | Section 3.4.1 |
| 10h | MEMPDPWRCTRL0 | Memory Power Domain Control Register 0 | Section 3.4.2 |
| 20h | PDCLKDIS | Power Domain Clock Disable Register | Section 3.4.3 |
| 24h | PDCLKDISSET | Power Domain Clock Disable Set Register | Section 3.4.4 |
| 28h | PDCLKDISCLR | Power Domain Clock Disable Clear Register | Section 3.4.5 |
| 40h | LOGICPDPWRSTAT0 | Logic Power Domain PD2 Power Status Register | Section 3.4.6 |
| 44h | LOGICPDPWRSTAT1 | Logic Power Domain PD3 Power Status Register | Section 3.4.7 |
| 48h | LOGICPDPWRSTAT2 | Logic Power Domain PD4 Power Status Register | Section 3.4.8 |
| 4Ch | LOGICPDPWRSTAT3 | Logic Power Domain PD5 Power Status Register | Section 3.4.9 |
| 80h | MEMPDPWRSTAT0 | Memory Power Domain RAM_PD1 Power Status Register | Section 3.4.10 |
| 84h | MEMPDPWRSTAT1 | Memory Power Domain RAM_PD2 Power Status Register | Section 3.4.11 |
| A0h | GLOBALCTRL1 | Global Control Register 1 | Section 3.4.12 |
| A8h | GLOBALSTAT | Global Status Register | Section 3.4.13 |
| ACh | PRCKEYREG | PSCON Diagnostic Compare Key Register | Section 3.4.14 |
| B0h | LPDDCSTAT1 | LogicPD PSCON Diagnostic Compare Status Register 1 | Section 3.4.15 |
| B4h | LPDDCSTAT2 | LogicPD PSCON Diagnostic Compare Status Register 2 | Section 3.4.16 |
| B8h | MPDDCSTAT1 | Memory PD PSCON Diagnostic Compare Status Register 1 | Section 3.4.17 |
| BCh | MPDDCSTAT2 | Memory PD PSCON Diagnostic Compare Status Register 2 | Section 3.4.18 |
| C0h | ISODIAGSTAT | Isolation Diagnostic Status Register | Section 3.4.19 |

### 3.4.1 LOGICPDPWRCTRL0 Register (Offset = 0h) [reset = X]

LOGICPDPWRCTRL0 is shown in Figure 3-3 and described in Table 3-2.

The default values of the control fields are determined by the device reset configuration word stored in the TI-OTP region of flash bank 0.

**Figure 3-3. LOGICPDPWRCTRL0 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RESERVED | | | | LOGICPDON0 | | | | RESERVED | | | | LOGICPDON1 | | | |
| R/W-0h | | | | R/WP-X | | | | R-0h | | | | R/WP-X | | | |
| **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |
| RESERVED | | | | LOGICPDON2 | | | | RESERVED | | | | LOGICPDON3 | | | |
| R-0h | | | | R/WP-X | | | | R-0h | | | | R/WP-X | | | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-2. LOGICPDPWRCTRL0 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-28 | RESERVED | R/W | 0h | Reads return 0. Writes have no effect. |
| 27-24 | LOGICPDON0 | R/WP | X | Read in User and Privileged Mode. Write in Privileged Mode only.<br><br>Read any other value: Power domain PD2 is in Active state.<br><br>Write any other value: Power domain PD2 is commanded to switch to Active state.<br>9h = Reserved.<br>Ah (W) = Power domain PD2 is commanded to switch to OFF state.<br>Ah (R) = Power domain PD2 is in OFF state. |
| 23-20 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 19-16 | LOGICPDON1 | R/WP | X | Read in User and Privileged Mode. Write in Privileged Mode only.<br><br>Read any other value: Power domain PD3 is in Active state.<br><br>Write any other value: Power domain PD3 is commanded to switch to Active state.<br>9h = Reserved.<br>Ah (W) = Power domain PD3 is commanded to switch to OFF state.<br>Ah (R) = Power domain PD3 is in OFF state. |
| 15-12 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 11-8 | LOGICPDON2 | R/WP | X | Read in User and Privileged Mode. Write in Privileged Mode only.<br><br>Read any other value: Power domain PD4 is in Active state.<br><br>Write any other value: Power domain PD4 is commanded to switch to Active state.<br>9h = Reserved.<br>Ah (W) = Power domain PD4 is commanded to switch to OFF state.<br>Ah (R) = Power domain PD4 is in OFF state. |
| 7-4 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |

**Table 3-2. LOGICPDPWRCTRL0 Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 3-0 | LOGICPDON3 | R/WP | X | Read in User and Privileged Mode. Write in Privileged Mode only.<br>Read any other value: Power domain PD5 is in Active state.<br>Write any other value: Power domain PD5 is commanded to switch to Active state.<br>9h = Reserved.<br>Ah (W) = Power domain PD5 is commanded to switch to OFF state.<br>Ah (R) = Power domain PD5 is in OFF state. |

### 3.4.2 MEMPDPWRCTRL0 Register (Offset = 10h) [reset = X]

MEMPDPWRCTRL0 is shown in Figure 3-4 and described in Table 3-3.

The default values of the control fields are determined by the device reset configuration word stored in the TI-OTP region of flash bank 0.

**Figure 3-4. MEMPDPWRCTRL0 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RESERVED | | | | MEMPDON0 | | | | RESERVED | | | | MEMPDON1 | | | |
| R-0h | | | | R/WP-X | | | | R-0h | | | | R/WP-X | | | |

| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | | | | | | | | | |
| R-0h | | | | | | | | | | | | | | | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-3. MEMPDPWRCTRL0 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-28 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 27-24 | MEMPDON0 | R/WP | X | Read in User and Privileged Mode. Write in Privileged Mode only.<br><br>Read any other value: Power domain RAM_PD1 is in Active state.<br><br>Write any other value: Power domain RAM_PD1 is commanded to switch to Active state.<br>9h = Reserved.<br>Ah (W) = Power domain RAM_PD1 is commanded to switch to OFF state.<br>Ah (R) = Power domain RAM_PD1 is in OFF state. |
| 23-20 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 19-16 | MEMPDON1 | R/WP | X | Read in User and Privileged Mode. Write in Privileged Mode only.<br><br>Read any other value: Power domain RAM_PD2 is in Active state.<br><br>Write any other value: Power domain RAM_PD2 is commanded to switch to Active state.<br>9h = Reserved.<br>Ah (W) = Power domain RAM_PD2 is commanded to switch to OFF state.<br>Ah (R) = Power domain RAM_PD2 is in OFF state. |
| 15-0 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |

### 3.4.3 PDCLKDIS Register (Offset = 20h) [reset = X]

PDCLKDIS is shown in Figure 3-5 and described in Table 3-4.

The default values of the control fields are determined by the device reset configuration word stored in the TI-OTP region of flash bank 0.

**Figure 3-5. PDCLKDIS Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** |
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** |
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |
| RESERVED | | | | PDCLK_ DIS_3 | PDCLK_ DIS_2 | PDCLK_ DIS_1 | PDCLK_ DIS_0 |
| R-0h | | | | R/WP-X | R/WP-X | R/WP-X | R/WP-X |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-4. PDCLKDIS Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-4 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 3 | PDCLK_DIS_3 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DIS[3]. Write in Privileged Mode only.<br>0h = Enable clocks to logic power domain PD5.<br>1h = Disable clocks to logic power domain PD5. |
| 2 | PDCLK_DIS_2 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DIS[2]. Write in Privileged Mode only.<br>0h = Enable clocks to logic power domain PD4.<br>1h = Disable clocks to logic power domain PD4. |
| 1 | PDCLK_DIS_1 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DIS[1]. Write in Privileged Mode only.<br>0h = Enable clocks to logic power domain PD3.<br>1h = Disable clocks to logic power domain PD3. |
| 0 | PDCLK_DIS_0 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DIS[0]. Write in Privileged Mode only.<br>0h = Enable clocks to logic power domain PD2.<br>1h = Disable clocks to logic power domain PD2. |

### 3.4.4 PDCLKDISSET Register (Offset = 24h) [reset = X]

PDCLKDISSET is shown in Figure 3-6 and described in Table 3-5.

The default values of the control fields are determined by the device reset configuration word stored in the TI-OTP region of flash bank 0.

**Figure 3-6. PDCLKDISSET Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** |
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** |
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |
| RESERVED | | | | PDCLK_ DISSET_3 | PDCLK_ DISSET_2 | PDCLK_ DISSET_1 | PDCLK_ DISSET_0 |
| R-0h | | | | R/WP-X | R/WP-X | R/WP-X | R/WP-X |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-5. PDCLKDISSET Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-4 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 3 | PDCLK_DISSET_3 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DISSET[3]. Write in Privileged Mode only.<br>0h = No effect to state of clocks to power domain PD5.<br>1h = Disable clocks to logic power domain PD5. |
| 2 | PDCLK_DISSET_2 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DISSET[2]. Write in Privileged Mode only.<br>0h = No effect to state of clocks to power domain PD4.<br>1h = Disable clocks to logic power domain PD4. |
| 1 | PDCLK_DISSET_1 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DISSET[1]. Write in Privileged Mode only.<br>0h = No effect to state of clocks to power domain PD3.<br>1h = Disable clocks to logic power domain PD3. |
| 0 | PDCLK_DISSET_0 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DISSET[0]. Write in Privileged Mode only.<br>0h = No effect to state of clocks to power domain PD2.<br>1h = Disable clocks to logic power domain PD2. |

### 3.4.5 PDCLKDISCLR Register (Offset = 28h) [reset = X]

PDCLKDISCLR is shown in Figure 3-7 and described in Table 3-6.

The default values of the control fields are determined by the device reset configuration word stored in the TI-OTP region of flash bank 0.

**Figure 3-7. PDCLKDISCLR Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| RESERVED | | | | PDCLK_ DISCLR_3 | PDCLK_ DISCLR_2 | PDCLK_ DISCLR_1 | PDCLK_ DISCLR_0 |
| R-0h | | | | R/WP-X | R/WP-X | R/WP-X | R/WP-X |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-6. PDCLKDISCLR Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-4 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 3 | PDCLK_DISCLR_3 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DIS[3]. Write in Privileged Mode only.<br>0h = No effect to state of clocks to power domain PD5.<br>1h = Enable clocks to logic power domain PD5. |
| 2 | PDCLK_DISCLR_2 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DIS[2]. Write in Privileged Mode only.<br>0h = No effect to state of clocks to power domain PD4.<br>1h = Enable clocks to logic power domain PD4. |
| 1 | PDCLK_DISCLR_1 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DIS[1]. Write in Privileged Mode only.<br>0h = No effect to state of clocks to power domain PD3.<br>1h = Enable clocks to logic power domain PD3. |
| 0 | PDCLK_DISCLR_0 | R/WP | X | Read in User and Privileged Mode returns the current value of PDCLK_DIS[0]. Write in Privileged Mode only.<br>0h = No effect to state of clocks to power domain PD2.<br>1h = Enable clocks to logic power domain PD2. |

### 3.4.6 LOGICPDPWRSTAT0 Register (Offset = 40h) [reset = X]

LOGICPDPWRSTAT0 is shown in Figure 3-8 and described in Table 3-7.

This is a read-only register. All writes are ignored.

The default values of the control fields are determined by the device reset configuration word stored in the TI-OTP region of flash bank 0.

**Figure 3-8. LOGICPDPWRSTAT0 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | LOGIC_IN_TRANS0 |
| R-0h | | | | | | | R-X |
| 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| RESERVED | | | | | | | MEM_IN_TRANS0 |
| R-0h | | | | | | | R-X |
| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| RESERVED | | | | | | | DOMAIN_ON0 |
| R-0h | | | | | | | R-X |
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| RESERVED | | | | | | LOGICPDPWR_STAT0 | |
| R-0h | | | | | | R-X | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-7. LOGICPDPWRSTAT0 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-25 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 24 | LOGIC_IN_TRANS0 | R | X | Logic in transition status for power domain PD2. Read in User and Privileged Mode.<br>0h = Logic in power domain PD2 is in the steady Active or OFF state.<br>1h = Logic in power domain PD2 is in the process of power-down/up. |
| 23-17 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 16 | MEM_IN_TRANS0 | R | X | Memory in transition status for power domain PD2. Read in User and Privileged Mode.<br>0h = Memory in power domain PD2 is in the steady Active or OFF state.<br>1h = Memory in power domain PD2 is in the process of power-down/up. |
| 15-9 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 8 | DOMAIN_ON0 | R | X | Current state of power domain PD2. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Power domain PD2 is in the OFF state.<br>1h = Power domain PD2 is in the Active state. |
| 7-2 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 1-0 | LOGICPDPWR_STAT0 | R | X | Logic power domain PD2 power state. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Logic power domain PD2 is switched OFF.<br>1h = Logic power domain PD2 is in Idle state.<br>2h = Reserved<br>3h = Logic power domain PD2 is in Active state. |

### 3.4.7 LOGICPDPWRSTAT1 Register (Offset = 44h) [reset = X]

LOGICPDPWRSTAT1 is shown in Figure 3-9 and described in Table 3-8.

This is a read-only register. All writes are ignored.

The default values of the control fields are determined by the device reset configuration word stored in the TI-OTP region of flash bank 0.

**Figure 3-9. LOGICPDPWRSTAT1 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | LOGIC_IN_TRANS1 |
| R-0h | | | | | | | R-X |
| 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| RESERVED | | | | | | | MEM_IN_TRANS1 |
| R-0h | | | | | | | R-X |
| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| RESERVED | | | | | | | DOMAIN_ON1 |
| R-0h | | | | | | | R-X |
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| RESERVED | | | | | | LOGICPDPWR_STAT1 | |
| R-0h | | | | | | R-X | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-8. LOGICPDPWRSTAT1 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-25 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 24 | LOGIC_IN_TRANS1 | R | X | Logic in transition status for power domain PD3. Read in User and Privileged Mode.<br>0h = Logic in power domain PD3 is in the steady Active or OFF state.<br>1h = Logic in power domain PD3 is in the process of power-down/up. |
| 23-17 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 16 | MEM_IN_TRANS1 | R | X | Memory in transition status for power domain PD3. Read in User and Privileged Mode.<br>0h = Memory in power domain PD3 is in the steady Active or OFF state.<br>1h = Memory in power domain PD3 is in the process of power-down/up. |
| 15-9 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 8 | DOMAIN_ON1 | R | X | Current state of power domain PD3. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Power domain PD3 is in the OFF state.<br>1h = Power domain PD3 is in the Active state. |
| 7-2 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 1-0 | LOGICPDPWR_STAT1 | R | X | Logic power domain PD3 power state. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Logic power domain PD3 is switched OFF.<br>1h = Logic power domain PD3 is in Idle state.<br>2h = Reserved<br>3h = Logic power domain PD3 is in Active state. |

### 3.4.8 LOGICPDPWRSTAT2 Register (Offset = 48h) [reset = X]

LOGICPDPWRSTAT2 is shown in Figure 3-10 and described in Table 3-9.

This is a read-only register. All writes are ignored.

The default values of the control fields are determined by the device reset configuration word stored in the TI-OTP region of flash bank 0.

**Figure 3-10. LOGICPDPWRSTAT2 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | LOGIC_IN_TRANS2 |
| R-0h | | | | | | | R-X |
| **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** |
| RESERVED | | | | | | | MEM_IN_TRANS2 |
| R-0h | | | | | | | R-X |
| **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** |
| RESERVED | | | | | | | DOMAIN_ON2 |
| R-0h | | | | | | | R-X |
| **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |
| RESERVED | | | | | | LOGICPDPWR_STAT2 | |
| R-0h | | | | | | R-X | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-9. LOGICPDPWRSTAT2 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-25 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 24 | LOGIC_IN_TRANS2 | R | X | Logic in transition status for power domain PD4. Read in User and Privileged Mode.<br>0h = Logic in power domain PD4 is in the steady Active or OFF state.<br>1h = Logic in power domain PD4 is in the process of power-down/up. |
| 23-17 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 16 | MEM_IN_TRANS2 | R | X | Memory in transition status for power domain PD4. Read in User and Privileged Mode.<br>0h = Memory in power domain PD4 is in the steady Active or OFF state.<br>1h = Memory in power domain PD4 is in the process of power-down/up. |
| 15-9 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 8 | DOMAIN_ON2 | R | X | Current state of power domain PD4. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Power domain PD4 is in the OFF state.<br>1h = Power domain PD4 is in the Active state. |
| 7-2 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 1-0 | LOGICPDPWR_STAT2 | R | X | Logic power domain PD4 power state. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Logic power domain PD4 is switched OFF.<br>1h = Logic power domain PD4 is in Idle state.<br>2h = Reserved<br>3h = Logic power domain PD4 is in Active state. |

### 3.4.9 LOGICPDPWRSTAT3 Register (Offset = 4Ch) [reset = X]

LOGICPDPWRSTAT3 is shown in Figure 3-11 and described in Table 3-10.

This is a read-only register. All writes are ignored.

The default values of the control fields are determined by the device reset configuration word stored in the TI-OTP region of flash bank 0.

**Figure 3-11. LOGICPDPWRSTAT3 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | LOGIC_IN_TRANS3 |
| R-0h | | | | | | | R-X |
| 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| RESERVED | | | | | | | MEM_IN_TRANS3 |
| R-0h | | | | | | | R-X |
| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| RESERVED | | | | | | | DOMAIN_ON3 |
| R-0h | | | | | | | R-X |
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| RESERVED | | | | | | LOGICPDPWR_STAT3 | |
| R-0h | | | | | | R-X | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-10. LOGICPDPWRSTAT3 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-25 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 24 | LOGIC_IN_TRANS3 | R | X | Logic in transition status for power domain PD5. Read in User and Privileged Mode.<br>0h = Logic in power domain PD5 is in the steady Active or OFF state.<br>1h = Logic in power domain PD5 is in the process of power-down/up. |
| 23-17 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 16 | MEM_IN_TRANS3 | R | X | Memory in transition status for power domain PD5. Read in User and Privileged Mode.<br>0h = Memory in power domain PD5 is in the steady Active or OFF state.<br>1h = Memory in power domain PD5 is in the process of power-down/up. |
| 15-9 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 8 | DOMAIN_ON3 | R | X | Current state of power domain PD5. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Power domain PD5 is in the OFF state.<br>1h = Power domain PD5 is in the Active state. |
| 7-2 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 1-0 | LOGICPDPWR_STAT3 | R | X | Logic power domain PD5 power state. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Logic power domain PD5 is switched OFF.<br>1h = Logic power domain PD5 is in Idle state.<br>2h = Reserved<br>3h = Logic power domain PD5 is in Active state. |

### 3.4.10 MEMPDPWRSTAT0 Register (Offset = 80h) [reset = X]

MEMPDPWRSTAT0 is shown in Figure 3-12 and described in Table 3-11.

This is a read-only register. All writes are ignored.

The default values of the control fields are determined by the device reset configuration word stored in the TI-OTP region of flash bank 0.

**Figure 3-12. MEMPDPWRSTAT0 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | LOGIC_IN_TRANS0 |
| R-0h | | | | | | | R-X |
| 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| RESERVED | | | | | | | MEM_IN_TRANS0 |
| R-0h | | | | | | | R-X |
| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| RESERVED | | | | | | | DOMAIN_ON0 |
| R-0h | | | | | | | R-X |
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| RESERVED | | | | | | MEMPDPWR_STAT0 | |
| R-0h | | | | | | R-X | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-11. MEMPDPWRSTAT0 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-25 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 24 | LOGIC_IN_TRANS0 | R | X | Logic in transition status for power domain RAM_PD1. This power domain only contains SRAM macros. However, an SRAM macro also has some digital logic controlled by the PSCON. Therefore, a memory power domain also contains a logic status indicator. Read in User and Privileged Mode.<br>0h = Logic in power domain RAM_PD1 is in the steady Active or OFF state.<br>1h = Logic in power domain RAM_PD1 is in the process of power-down/up. |
| 23-17 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 16 | MEM_IN_TRANS0 | R | X | Memory in transition status for power domain RAM_PD1. Read in User and Privileged Mode.<br>0h = Memory in power domain RAM_PD1 is in the steady Active or OFF state.<br>1h = Memory in power domain RAM_PD1 is in the process of power-down/up. |
| 15-9 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 8 | DOMAIN_ON0 | R | X | Current state of power domain RAM_PD1. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Power domain RAM_PD1 is in the OFF state.<br>1h = Power domain RAM_PD1 is in the Active state. |
| 7-2 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |

**Table 3-11. MEMPDPWRSTAT0 Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 1-0 | MEMPDPWR_STAT0 | R | X | Memory power domain RAM_PD1 power state. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Memory power domain RAM_PD1 is switched OFF.<br>1h = Reserved<br>2h = Reserved<br>3h = Logic power domain RAM_PD1 is in Active state. |

### *3.4.11 MEMPDPWRSTAT1 Register (Offset = 84h) [reset = X]*

MEMPDPWRSTAT1 is shown in Figure 3-13 and described in Table 3-12.

This is a read-only register. All writes are ignored.

The default values of the control fields are determined by the device reset configuration word stored in the TI-OTP region of flash bank 0.

**Figure 3-13. MEMPDPWRSTAT1 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | LOGIC_IN_TRANS1 |
| R-0h | | | | | | | R-X |
| **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** |
| RESERVED | | | | | | | MEM_IN_TRANS1 |
| R-0h | | | | | | | R-X |
| **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** |
| RESERVED | | | | | | | DOMAIN_ON1 |
| R-0h | | | | | | | R-X |
| **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |
| RESERVED | | | | | | MEMPDPWR_STAT1 | |
| R-0h | | | | | | R-X | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-12. MEMPDPWRSTAT1 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-25 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 24 | LOGIC_IN_TRANS1 | R | X | Logic in transition status for power domain RAM_PD2. This power domain only contains SRAM macros. However, an SRAM macro also has some digital logic controlled by the PSCON. Therefore, a memory power domain also contains a logic status indicator. Read in User and Privileged Mode.<br>0h = Logic in power domain RAM_PD2 is in the steady Active or OFF state.<br>1h = Logic in power domain RAM_PD2 is in the process of power-down/up. |
| 23-17 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 16 | MEM_IN_TRANS1 | R | X | Memory in transition status for power domain RAM_PD2. Read in User and Privileged Mode.<br>0h = Memory in power domain RAM_PD2 is in the steady Active or OFF state.<br>1h = Memory in power domain RAM_PD2 is in the process of power-down/up. |
| 15-9 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 8 | DOMAIN_ON1 | R | X | Current state of power domain RAM_PD2. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Power domain RAM_PD2 is in the OFF state.<br>1h = Power domain RAM_PD2 is in the Active state. |
| 7-2 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |

**Table 3-12. MEMPDPWRSTAT1 Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 1-0 | MEMPDPWR_STAT1 | R | X | Memory power domain RAM_PD2power state. The default value of this field is controlled by the device reset configuration word in the TI-OTP region of flash bank 0. Read in User and Privileged Mode.<br>0h = Memory power domain RAM_PD2 is switched OFF.<br>1h = Reserved<br>2h = Reserved<br>3h = Logic power domain RAM_PD2 is in Active state. |

### 3.4.12 GLOBALCTRL1 Register (Offset = A0h) [reset = 0h]

GLOBALCTRL1 is shown in Figure 3-14 and described in Table 3-13.

**Figure 3-14. GLOBALCTRL1 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** |
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** |
| RESERVED | | | | | | | PMCTRL_ PWRDN |
| R-0h | | | | | | | R/WP-0h |
| **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |
| RESERVED | | | | | | | AUTO_CLK_ WAKE_ENA |
| R-0h | | | | | | | R/WP-0h |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-13. GLOBALCTRL1 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-9 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 8 | PMCTRL_PWRDN | R/WP | 0h | PMC/PSCON Power Down. Read in User and Privileged Mode returns current value of PMCTRL PWRDN. Write in Privileged mode only.<br>0h = Enable the clock to pmctrl_wakeup block.<br>1h = Disable the clock to pmctrl_wakeup block, which contains PMC and all PSCONs. |
| 7-1 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 0 | AUTO_CLK_WAKE_ENA | R/WP | 0h | Automatic Clock Enable on Wake Up. Read in User and Privileged Mode returns current value of AUTO CLK WAKE ENA. Write in Privileged mode only.<br>0h = Disable automatic clock wake up. The application must enable clocks by clearing the correct bit in the PDCLK_DIS register.<br>1h = Enable automatic clock wake up when a power domain transitions to Active state. |

### 3.4.13 GLOBALSTAT Register (Offset = A8h) [reset = 1h]

GLOBALSTAT is shown in Figure 3-15 and described in Table 3-14.

**Figure 3-15. GLOBALSTAT Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** |
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** |
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |
| RESERVED | | | | | | | PMCTRL_IDLE |
| R-0h | | | | | | | R-1h |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-14. GLOBALSTAT Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-1 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 0 | PMCTRL_IDLE | R | 1h | State of PMC and all PSCONs. The PMM captures the status of PMC and PSCONs as they do not have a register interface to the host CPU.<br>0h = PMC and PSCONs for all power domains are in the process of generating power state transition control sequence for logic and/or SRAM.<br>1h = PMC and PSCONs for all power domains have completed generating power state transition control sequence triggered by PMC input control signals. |

### 3.4.14 PRCKEYREG Register (Offset = ACh) [reset = 0h]

PRCKEYREG is shown in Figure 3-16 and described in Table 3-15.

**Figure 3-16. PRCKEYREG Register**

| 31–4 | 3–0 |
|---|---|
| RESERVED | MKEY |
| R-0h | R/WP-0h |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; *-n* = value after reset

**Table 3-15. PRCKEYREG Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-4 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 3-0 | MKEY | R/WP | 0h | Diagnostic PSCON Mode Key. The mode key is applied to all individual PSCON compare units. Read in User and Privileged mode returns the current value of MKEY. Write in Privileged mode only.<br>0h = Lock Step mode<br>6h = Self-test mode<br>9h = Error Forcing mode<br>Fh = Self-test Error Forcing Mode |

### 3.4.15 LPDDCSTAT1 Register (Offset = B0h) [reset = 0h]

LPDDCSTAT1 is shown in Figure 3-17 and described in Table 3-16.

**Figure 3-17. LPDDCSTAT1 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | | | | | | LCMPE_3_0 | | | |
| R-0h | | | | | | | | | | | | R/W1CP-0h | | | |
| **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |
| RESERVED | | | | | | | | | | | | LSTC_3_0 | | | |
| R-0h | | | | | | | | | | | | R-0h | | | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-16. LPDDCSTAT1 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-20 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 19-16 | LCMPE_3_0 | R/W1CP | 0h | Logic Power Domain Compare Error. Each of these bits corresponds to a logic power domain:<br>Bit 3 for PD5<br>Bit 2 for PD4<br>Bit 1 for PD3<br>Bit 0 for PD2<br><br>Read in User and Privileged Mode. Write in Privileged mode only.<br>0h (W) = Writing 0 has no effect.<br>0h (R) = PSCON signals are identical.<br>1h (W) = Clears the corresponding LCMPE[n] bit, if set.<br>1h (R) = PSCON signal compare mismatch identified. |
| 15-4 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 3-0 | LSTC_3_0 | R | 0h | Logic Power Domain Self-test Complete. Each of these bits corresponds to a logic power domain:<br>Bit 3 for PD5<br>Bit 2 for PD4<br>Bit 1 for PD3<br>Bit 0 for PD2<br><br>Read in User and Privileged Mode. Writes have no effect.<br>0h = Self-test is ongoing if self-test mode is entered.<br>1h = Self-test is complete. |

### 3.4.16 LPDDCSTAT2 Register (Offset = B4h) [reset = 0h]

LPDDCSTAT2 is shown in Figure 3-18 and described in Table 3-17.

**Figure 3-18. LPDDCSTAT2 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | | | | | | LSTET_3_0 | | | |
| R-0h | | | | | | | | | | | | R-0h | | | |
| **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |
| RESERVED | | | | | | | | | | | | LSTE_3_0 | | | |
| R-0h | | | | | | | | | | | | R-0h | | | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; *-n* = value after reset

**Table 3-17. LPDDCSTAT2 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-20 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 19-16 | LSTET_3_0 | R | 0h | Logic Power Domain Self-test Error Type. Each of these bits corresponds to a logic power domain:<br>Bit 3 for PD5<br>Bit 2 for PD4<br>Bit 1 for PD3<br>Bit 0 for PD2<br><br>Read in User and Privileged Mode. Writes have no effect.<br>0h = Self-test failed during compare match test.<br>1h = Self-test failed during compare mismatch test. |
| 15-4 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 3-0 | LSTE_3_0 | R | 0h | Logic Power Domain Self-test Error. Each of these bits corresponds to a logic power domain:<br>Bit 3 for PD5<br>Bit 2 for PD4<br>Bit 1 for PD3<br>Bit 0 for PD2<br><br>Read in User and Privileged Mode. Writes have no effect.<br>0h = Self-test passed.<br>1h = Self-test failed. |

### 3.4.17 MPDDCSTAT1 Register (Offset = B8h) [reset = 0h]

MPDDCSTAT1 is shown in Figure 3-19 and described in Table 3-18.

This register shows the interrupt status (before enabling) and allows setting of the interrupt status.

**Figure 3-19. MPDDCSTAT1 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** |
| RESERVED | | | | | | MCMPE_1_0 | |
| R-0h | | | | | | R/W1CP-0h | |
| **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** |
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |
| RESERVED | | | | | | MSTC_1_0 | |
| R-0h | | | | | | R-0h | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; *-n* = value after reset

**Table 3-18. MPDDCSTAT1 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-18 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 17-16 | MCMPE_1_0 | R/W1CP | 0h | Memory Power Domain Compare Error. Each of these bits corresponds to a memory power domain:<br>Bit 1 for RAM_PD2<br>Bit 0 for RAM_PD1<br><br>Read in User and Privileged Mode. Write in Privileged mode only.<br>0h (W) = Writing 0 has no effect.<br>0h (R) = PSCON signals are identical.<br>1h (W) = Clears the corresponding MCMPE[n] bit, if set.<br>1h (R) = PSCON signal compare mismatch identified. |
| 15-2 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 1-0 | MSTC_1_0 | R | 0h | Memory Power Domain Self-test Complete. Each of these bits corresponds to a memory power domain:<br>Bit 1 for RAM_PD2<br>Bit 0 for RAM_PD1<br><br>Read in User and Privileged Mode. Writes have no effect.<br>0h = Self-test is ongoing if self-test mode is entered.<br>1h = Self-test is complete. |

### 3.4.18 MPDDCSTAT2 Register (Offset = BCh) [reset = 0h]

MPDDCSTAT2 is shown in Figure 3-20 and described in Table 3-19.

**Figure 3-20. MPDDCSTAT2 Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| RESERVED | | | | | | MSTET_1_0 | |
| R-0h | | | | | | R-0h | |
| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| RESERVED | | | | | | | |
| R-0h | | | | | | | |
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| RESERVED | | | | | | MSTE_1_0 | |
| R-0h | | | | | | R-0h | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-19. MPDDCSTAT2 Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-18 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 17-16 | MSTET_1_0 | R | 0h | Memory Power Domain Self-test Error Type. Each of these bits corresponds to a memory power domain:<br>Bit 1 for RAM_PD2<br>Bit 0 for RAM_PD1<br><br>Read in User and Privileged Mode. Writes have no effect.<br>0h = Self-test failed during compare match test.<br>1h = Self-test failed during compare mismatch test. |
| 15-2 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 1-0 | MSTE_1_0 | R | 0h | Memory Power Domain Self-test Error. Each of these bits corresponds to a memory power domain:<br>Bit 1 for RAM_PD2<br>Bit 0 for RAM_PD1<br><br>Read in User and Privileged Mode. Writes have no effect.<br>0h = Self-test passed.<br>1h = Self-test failed. |

### 3.4.19 ISODIAGSTAT Register (Offset = C0h) [reset = 0h]

ISODIAGSTAT is shown in Figure 3-21 and described in Table 3-20.

There is an ISO_DIAG cell implemented separately for each logic power domain. This is a special tie-off cell that reads a value of 1 when the logic power domain is powered up. This cell has an isolation value of 0. That is, when this logic power domain is turned off, this cell will read 0. The ISO_DIAG statuses for each logic power domain is reflected in the ISODIAGSTAT register. The application can poll this diagnostic register to make sure that a domain that has been commanded to turn off has actually received the command.

**Figure 3-21. ISODIAGSTAT Register**

| 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | | | | | | | | | |
| R-0h | | | | | | | | | | | | | | | |

| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RESERVED | | | | | | | | | | | | ISO_DIAG_3_0 | | | |
| R-0h | | | | | | | | | | | | R-0h | | | |

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -*n* = value after reset

**Table 3-20. ISODIAGSTAT Register Field Descriptions**

| Bit | Field | Type | Reset | Description |
|---|---|---|---|---|
| 31-4 | RESERVED | R | 0h | Reads return 0. Writes have no effect. |
| 3-0 | ISO_DIAG_3_0 | R | 0h | Isolation Diagnostic. Each of these bits corresponds to a logic power domain:<br>Bit 3 for PD5<br>Bit 2 for PD4<br>Bit 1 for PD3<br>Bit 0 for PD2<br><br>Read in User and Privileged Mode. Writes have no effect.<br>0h = Isolation is enabled for corresponding power domain.<br>1h = Isolation is disabled for corresponding power domain. |